

Class 12 Project  
Class 12 Project Proposal Form

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| Student name                                                                                                                                                                        | Louis Benne                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Central idea<br>(what is your project?)                                                                                                                                             | Build a multi-system computer server/home lab (10in rack) while developing IT skills and certifications in networking, hardware and software systems and setups, IoT, NAS, cybersecurity, LoRa (radio coms) & SDRs, and self-hosted applications, creating a personal sandbox lab.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Aims<br>(what do you plan to have learned/achieved/produced by the end of this project?)                                                                                            | <ul style="list-style-type: none"> <li>• A working multi-system home lab: server, network, and virtualisation environment</li> <li>• Practical knowledge of networking, hardware, and systems design and management</li> <li>• Possibly progress towards IT certifications (e.g. Network+, Security+, CCNA)</li> <li>• Working IoT devices integrated into the network</li> <li>• Experiment with LoRa (long-range low-power communication) and SDR (software-defined radio)</li> <li>• Self-hosted applications running in a sandbox environment</li> <li>• Cybersecurity skills</li> <li>• Hopefully explore a system (like LoRa) in-depth, then write a report for a LO in a subject in the Class 12 curriculum (like physics)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Outline of project<br>(inc. practical work, written work, resources, how you will keep track of your progress; + bonus, any ideas you have for your exhibition and/or presentation) | <p>Practical work:</p> <ul style="list-style-type: none"> <li>• Source and set up hardware (server, switches, access points, Raspberry Pis/SBCs)</li> <li>• Build a home lab network (VLANs, firewall, pfSense/OPNsense)</li> <li>• Install a hypervisor (Proxmox/ESXi) and run multiple VMs/containers</li> <li>• Set up self-hosted apps (e.g. Pi-hole, Nextcloud, Home Assistant, Jellyfin)</li> <li>• Experiment with LoRa and SDR hardware</li> <li>• Practise and explore cybersecurity using what I have made</li> </ul> <p>Written work:</p> <ul style="list-style-type: none"> <li>• Full report of the lab and its functions, what I did with it, and how. Write-up of software I will be running and network diagrams and documentation for each stage</li> <li>• Ideally a full GitHub page. Ex: <a href="https://mini-rack.jeffgeerling.com/">https://mini-rack.jeffgeerling.com/</a></li> <li>• If I choose to do a certification, I might do a main book with content of what I learnt.</li> </ul> <p>Resources:</p> <ul style="list-style-type: none"> <li>• Server/networking hardware, Raspberry Pis, LoRa modules, SDR</li> <li>• YouTube, TryHackMe, Hack the Box and the internet.</li> </ul> <p>Tracking progress:</p> <ul style="list-style-type: none"> <li>• Build log on GitHub which is updated regularly with photos, configs, and lessons learned, like a blog, and everything I do will be on my GitHub.</li> </ul> <p>Exhibition/presentation ideas:</p> <ul style="list-style-type: none"> <li>• Rack on show</li> <li>• Poster/diagram showing full network topology or Rack specs</li> <li>• Interactive demos with radios and LoRa visually.</li> </ul> |
| Outline of timeline                                                                                                                                                                 | <p>What I aim to have achieved by...</p> <p>September 2026: Source hardware, plan and start building. Possibly begin study for certs.</p> <p>December 2026: built core server and hypervisor running; basic self-hosted apps live</p> <p>January 2027: IoT devices integrated; begin cybersecurity hardening/testing</p> <p>Easter 2027: LoRa/SDR experiments underway; certifications completed</p> <p>May 2027: Final documentation complete, exhibition and presentations ready.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Supervisor<br>(Is there a teacher at the school whom you would like to request to supervise your project?)                                                                          | Dr Figgols? Don't mind. Whoever is vaguely familiar with what I'm doing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Other mentors<br>(names & contact details of any mentors outside school)                                                                                                            | The internet. And <a href="https://mini-rack.jeffgeerling.com/">https://mini-rack.jeffgeerling.com/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |